

Homework 9

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Question 1

a)

$$f_Y(y) = \int_0^y 2e^{-(x+y)} dx = -2e^{-(x+y)} \Big|_0^y = 2e^{-y} - 2e^{-2y}$$

As $e^{-y} > e^{-2y}$ this is a positive function.

$$\begin{aligned} \int_0^\infty (2e^{-y} - 2e^{-2y}) dy &= 2 \left(-e^{-y} + e^{-2y}/2 \right) \Big|_0^\infty \\ &= 2(1 - 1/2) = 1 \end{aligned}$$

So this is in fact a pdf.

b)

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f(y)} = \frac{2e^{-(x+y)}}{2e^{-y} - 2e^{-2y}} = \frac{e^{-x}}{1 - e^{-y}} I(0 < x < y < \infty)$$

Given $y > 0$ this will always be positive.

$$\int_0^y \frac{e^{-x}}{1 - e^{-y}} dx = \frac{1}{1 - e^{-y}} (-e^{-x}) \Big|_0^y = \frac{-e^{-y} + 1}{1 - e^{-y}} = 1$$

So this is indeed a pdf.

c)

First we need to find $f_X(x)$:

$$f_X(x) = \int_x^\infty 2e^{-(x+y)} dy = -2e^{-(x+y)} \Big|_x^\infty = 2e^{-2x}$$

Then:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f(y)} = \frac{2e^{-(x+y)}}{2e^{-2y}} I(0 < x < y < \infty) = \frac{e^{-y}}{e^{-2y}} I(0 < x < y < \infty)$$

Both the numerator and denominator are positive so this quantity is also positive.

$$\int_x^\infty \frac{e^{-y}}{e^{-2y}} dy = \frac{1}{e^{-x}} (-e^{-y}) \Big|_x^\infty = \frac{0 + e^{-x}}{e^{-x}} = 1$$

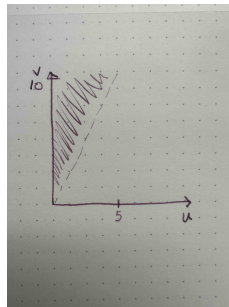
So this is indeed a pdf.

d)

X and Y are not independent as you cannot factor $I(0 < x < y < \infty)$ into components only of x and y .

Question 2

a)



b)

We have: $x = u$ and $y = v - u$. So:

$$\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = 1 - 0 = 1$$

Therefore:

$$f_{U,V}(u, v) = 2e^{-v} I(0 < u < \infty) I(2u < v < \infty)$$

c)

$$f_U(u) = \int_{2u}^{\infty} 2e^{-v} dv = 2(-e^{-v}) \Big|_0^{\infty} = 2e^{-2u}$$

$$f_V(v) = \int_0^{v/2} 2e^{-v} du = 2e^{-v}(u) \Big|_0^{v/2} = ve^{-v}$$

d)

Once again U and V are not independent because we cannot factor the I in terms of functions only of v and u .

Question 3

$f_X(x) = ce^{-x^2/a}$, $f_Y(y) = ce^{-y^2/a}$ and therefore as X and Y are independent:

$$f_{X,Y}(x,y) = c^2 e^{-(x^2+y^2)/a}$$

Now $y = v$ and $x = uv$ so that:

$$\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} v & u \\ 0 & 1 \end{bmatrix} = v$$

which in turn means that:

$$f_{U,V} = c^2 e^{-(u^2v^2+v^2)/a} |v| = c^2 e^{-v^2(u^2+1)/a} |v|$$

As this function is symmetric in v around 0 we have:

$$f_U(u) = 2 \int_0^{\infty} c^2 e^{-v^2(u^2+1)/a} v dv$$

$$= \frac{-c^2 a e^{-v^2(u^2+1)/a}}{(u^2+1)} \Big|_0^{\infty} = \frac{c^2 a}{(u^2+1)}$$

As $c = \sqrt{2\pi}$ and $a = 2$ this becomes:

$$f_U(u) = \frac{1}{\pi(u^2+1)}$$

which is the pdf for the Cauchy distribution.

Question 4

Let $V = X$. Then we have $x = v$ and $y = u - v$. This means that:

$$\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} = 0 - (1) = -1$$

This means that our $f(x, y)$ gets multiplied by 1 in the conversion to $f(u, v)$. Because our $f(x, y) = 2$ in all cases this means $f(u, v) = 2$ in all cases as well. To get $f_U(u)$ we therefore now just need to find the marginal distribution of U which will be all about understanding the bounds of integration.

a)

In this case, for a given u , v can range from 0 to u .

$$f_U(u) = \int_0^u 2dv = 2u$$

where u can range from $0 \rightarrow 1$. That is $f_U(u) = 2uI(0 \leq u \leq 1)$.

b)

In this case, for a given u , v can range from 0 to $u/2$ (as $x \leq y$).

$$f_U(u) = \int_0^{u/2} 2dv = u$$

In this case u can range from 0 to 2 so $f_U(u) = uI(0 \leq u \leq 2)$.

c)

In this case, if $u > 1$ then v can range from $u - 1 \rightarrow 1$. If $u < 1$ then v ranges from $0 \rightarrow u$.

That is:

$$f_U(u) = \begin{cases} \int_0^u dv = u & 0 \leq u \leq 1 \\ \int_{u-1}^1 dv = 2 - u & 1 < u \leq 2 \end{cases}$$

Question 5

a)

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{1}{1-\beta t}\right)^{\alpha_x} \left(\frac{1}{1-\beta t}\right)^{\alpha_y} = \left(\frac{1}{1-\beta t}\right)^{\alpha_x+\alpha_y}$$

So $X + Y \sim \text{Gamma}(\alpha_x + \alpha_y, \beta)$

b)

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\theta(e^t-1)}e^{\lambda(e^t-1)} = e^{(\theta+\lambda)(e^t-1)}$$

So $X + Y \sim \text{Poisson}(\theta + \lambda)$

Question 6

In this case we have a hierarchical model $Z|N$.

$$\mathbb{E}(Z) = \mathbb{E}(\mathbb{E}(Z|N))$$

$$\mathbb{E}(Z|N) = \mathbb{E}\left(\sum_{i=1}^N Y_i\right) = \sum_{i=1}^N \mathbb{E}(Y_i) = 10N$$

$$\mathbb{E}(Z) = \mathbb{E}(\mathbb{E}(Z|N)) = \sum_{N=1}^{13} \frac{1}{13} 10N = 70$$

Now onto the variance.

$$\text{Var}(Z) = \text{Var}(\mathbb{E}(Z|N)) + \mathbb{E}(\text{Var}(Z|N))$$

Let's start with the first of those two terms:

$$\begin{aligned} \text{Var}(\mathbb{E}(Z|N)) &= \mathbb{E}(\mathbb{E}(Z|N)^2) - \mathbb{E}(\mathbb{E}(Z|N))^2 \\ &= \sum_{N=1}^{13} \frac{1}{13} (10N)^2 - 70^2 = 6300 - 4900 = 1400 \end{aligned}$$

And then the second of the two terms:

$$\text{Var}(Z|N) = \text{Var}\left(\sum_{i=1}^N Y_i\right) = \sum_{i=1}^N \text{Var}(Y_i) = 10N$$

$$\mathbb{E}(\text{Var}(Z|N)) = \sum_{N=1}^{13} \frac{1}{13} 10N = 70$$

Therefore:

$$\text{Var}(Z) = \text{Var}(\mathbb{E}(Z|N)) + \mathbb{E}(\text{Var}(Z|N)) = 1400 + 70 = 1470$$

Question 7

$$P(Y < y) = P(WX < y)$$

By independence of W and X we have:

$$P(WX < y) = P(X < y)P(W = 1) + P(X > -y)P(W = -1)$$

Given $N(0, 1)$ is symmetric around 0, $P(X > -y) = P(X < y)$. Therefore we have:

$$P(Y < y) = P(X < y) (P(W = 1) + P(W = -1)) = P(X < y)$$

As X and Y share the same cdf and support they must also share the same pdf.

Question 8

a)

Let $\mu_Y = \mathbb{E}(Y)$, $\mu_{X_1} = \mathbb{E}(X_1)$, and $\mu_{X_2} = \mathbb{E}(X_2)$ then:

$$\text{cov}(X_1 + X_2, Y) = \mathbb{E}((X_1 + X_2 - \mu_{X_1} - \mu_{X_2})(Y - \mu_Y))$$

by virtue of the fact that $\mathbb{E}(X_1 + X_2) = \mu_{X_1} + \mu_{X_2}$

$$\begin{aligned} (X_1 + X_2 - \mu_{X_1} - \mu_{X_2})(Y - \mu_Y) &= X_1Y - X_1\mu_Y + X_2Y - X_2\mu_Y \\ -(\mu_{X_1}Y - \mu_{X_1}\mu_Y + \mu_{X_2}Y - \mu_{X_2}\mu_Y) &= (X_1Y - X_1\mu_Y - (\mu_{X_1}Y - \mu_{X_1}\mu_Y)) \\ + (X_2Y - X_2\mu_Y - (\mu_{X_2}Y - \mu_{X_2}\mu_Y)) &= (X_1 - \mu_{X_1})(Y - \mu_Y) \end{aligned}$$

$$+(X_2 - \mu_{X_2})(Y - \mu_Y)$$

Therefore:

$$\begin{aligned}\text{cov}(X_1 + X_2, Y) &= \mathbb{E}((X_1 + X_2 - \mu_{X_1} - \mu_{X_2})(Y - \mu_Y)) \\ &= \mathbb{E}((X_1 - \mu_{X_1})(Y - \mu_Y) + (X_2 - \mu_{X_2})(Y - \mu_Y)) \\ &= \mathbb{E}((X_1 - \mu_{X_1})(Y - \mu_Y)) + \mathbb{E}((X_2 - \mu_{X_2})(Y - \mu_Y)) \\ &= \text{cov}(X_1, Y) + \text{cov}(X_2, Y)\end{aligned}$$

b)

$$\begin{aligned}\text{cov}(X, a) &= \mathbb{E}((X - \mathbb{E}(X))(a - \mathbb{E}(a))) \\ &= \mathbb{E}((X - \mathbb{E}(X))(a - a)) = \mathbb{E}(0) = 0\end{aligned}$$